

JEE MAIN REVISION MASTER: FUNCTIONS 2026 2

QUESTIONS 3.33% WEIGHTAGE ↓33.4%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Functions - Core Application	36 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Functions - Core Application (Very High Yield — 36 PYQs)

Core Principle: Key problem pattern in Kcet Functions.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $f(x)=(x-1)^2$ for $x \geq 1$, $g(x)$ is a function whose graph is the reflection of the graph of $f(x)$ in the line $y=x$, then $g(x)$ i..." [Ans: C]
- [2026 Shift] Q#2 (MCQ): "The domain of the function $\sqrt{(x-7 / 9-x)}$ is..." [Ans: B]
- [2025 Shift] Q#1 (MCQ): "Let the functions " f " and " g " be $f:[0,(\pi / 2)] \rightarrow R$ given by $f(x)=\sin x$ and $g:[0,(\pi / 2)] \rightarrow R$ given ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$f(x)=(x-1)^2$ for $x \geq 1$, $g(x)$ is a function whose graph is the reflection of the graph of $f(x)$ in the line $y=x$, then $g(x)$ is

- (A) $-\sqrt{x}+1$
- (B) $\sqrt{x}-1$
- (C) $\sqrt{x}+1$ [Correct]
- (D) $\sqrt{x-1}$

Correct Answer: C

Solution Analysis: The graph of $g(x)$ is the reflection of the graph of $f(x)$ in the line $y=x$. Reflection in the line $y=x$ gives the inverse relation of the function. Given, $f(x)=(x-1)^2, x \geq 1$. Let $y=(x-1)^2$. Since $x \geq 1$, we have $x-1 \geq 0$. So when we take square root, $\sqrt{y}=x-1$. Hence, $x=\sqrt{y}+1$. Now interchange x and y to get the reflected function: $y=\sqrt{x}+1$. So, $g(x)=\sqrt{x}+1$. Hence the correct option is Option C $\sqrt{x}+1$.